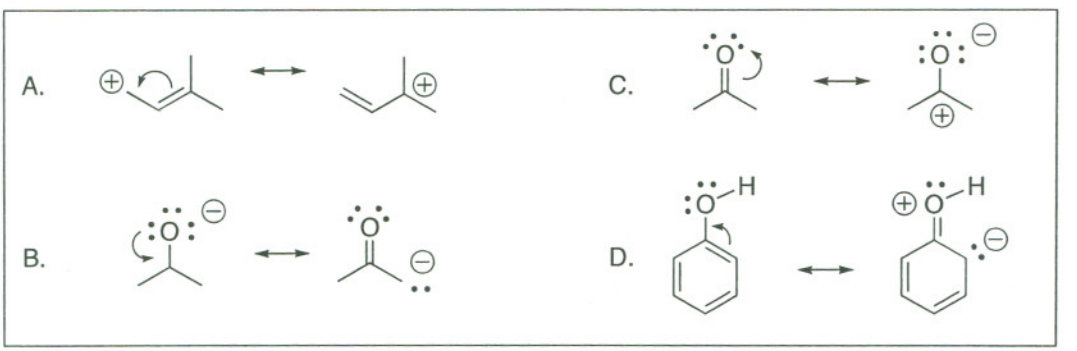
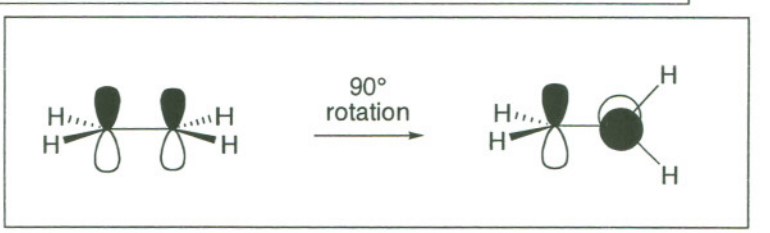
Organic Chemistry cumulative exam **March 3, 2006**

The following 10 questions were taken from the final exam for Honors/Majors Organic Chemistry I last year (total of 30 questions).

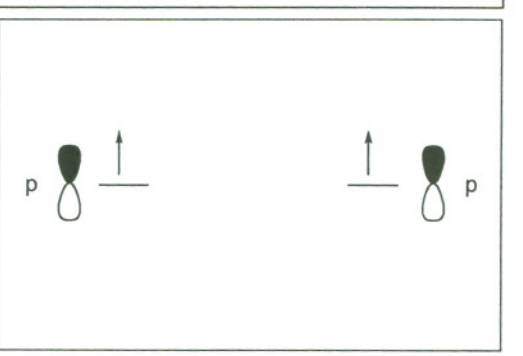
1. For each pair of structures below, indicate whether the electron pushing arrow shown will (Y) or will not (N) lead to the structures on the right.



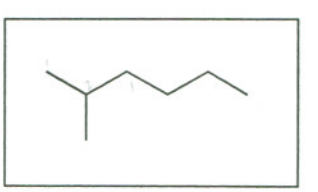
2. How much energy in kcal/mol does it take to perform the following bond rotation in ethene?



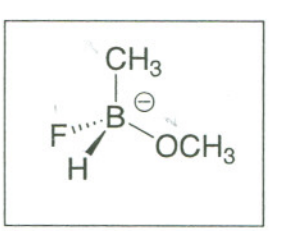
3. Draw the energy level diagram showing how two separate p-orbitals combine using the LCAO method. Draw the shape, indicate phasing and give the symbols of the new localized molecular orbitals.



4. Draw the Newman projection along the **C2-C3** bond of the alkane to the right in the lowest energy conformation (C2 in front).

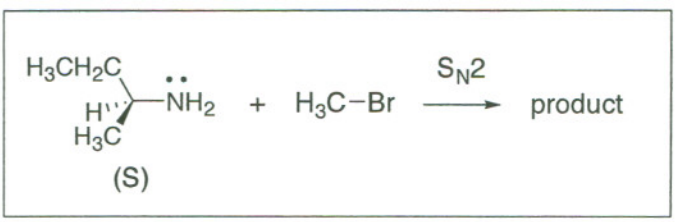


5. A. Is the borate to the right chiral? (Y or N).
B. If so, is it R or S?

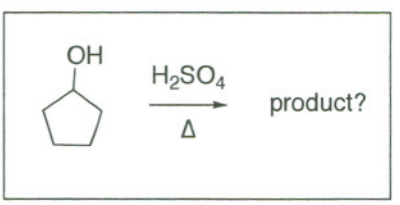


6. Using the S_N2 reaction of NH_3 and CH_3Br as an example, **briefly explain using structures** the theoretical basis (i.e. FMO theory) for the experimental fact that nucleophiles attack electrophiles opposite to the leaving group (backside attack).

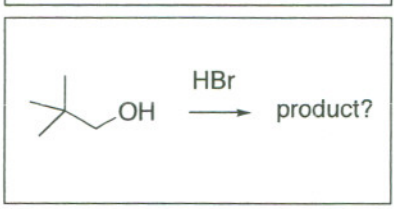
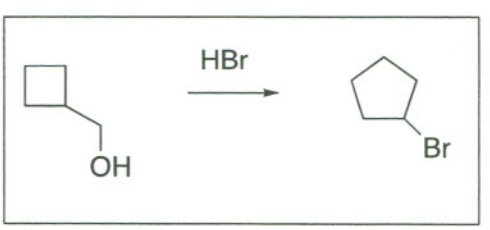
7. What is the **absolute configuration** of the product in the following S_N2 reaction?



8. Draw the product of the reaction to the right



9. Draw the product of the reaction to the right.



10. Draw the step-by-step electron pushing mechanism of the reaction to the right.
Hint: see previous problem.